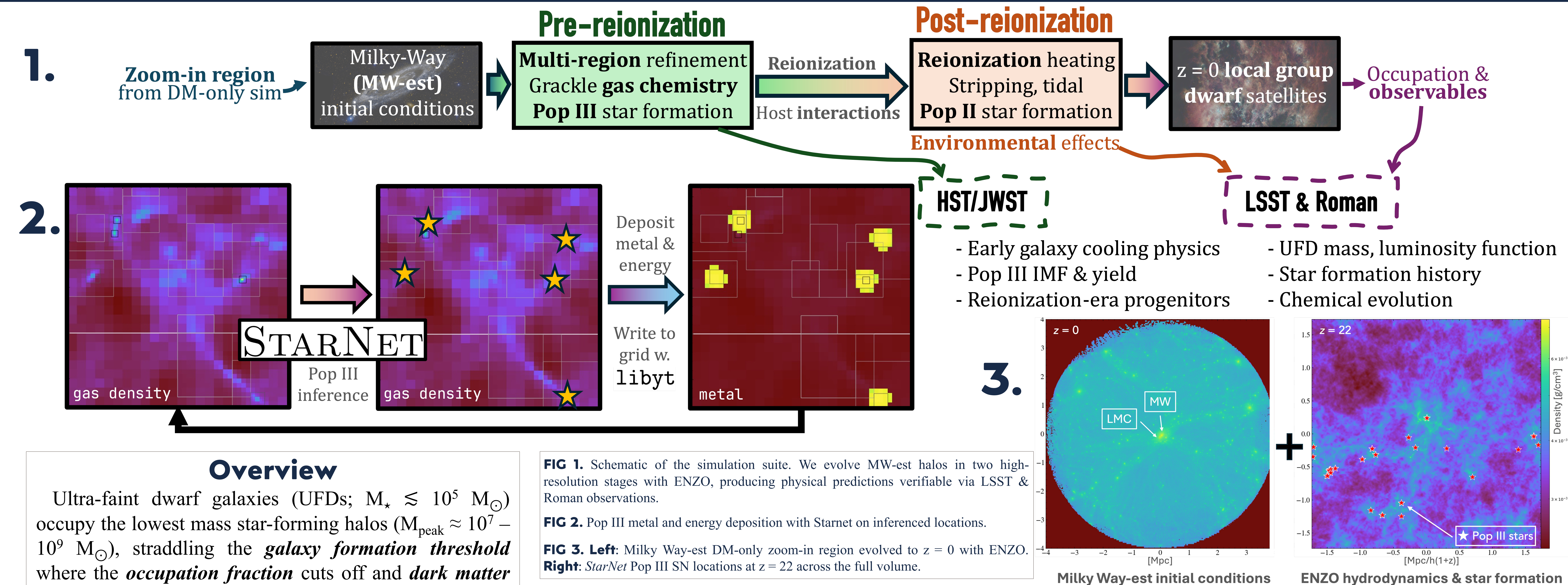


High-resolution Hydrodynamic Simulations of Ultra-Faint Milky Way Satellites

Zewei Wu¹

with Ethan Nadler¹, Sandip Roy^{1,2}, Anna Wright³, and Michael Norman^{1,2}

¹Department of Astronomy & Astrophysics, UCSD; ²San Diego Supercomputer Center; ³Center for Computational Astrophysics, Flatiron Institute



Overview

Ultra-faint dwarf galaxies (UFDs; $M_\star \lesssim 10^5 M_\odot$) occupy the lowest mass star-forming halos ($M_{\text{peak}} \approx 10^7 - 10^9 M_\odot$), straddling the *galaxy formation threshold* where the *occupation fraction* cuts off and *dark matter models* diverge most. Their resolved stellar populations constrains physics like H_2 cooling, Pop III stars, SNe feedback, reionization, and tidal evolution.

LSST will reveal up to ~ 100 resolved MW satellites; interpreting them requires simulations with resolution ($\sim 10 M_\odot$, $\sim \text{pc}$) to **resolve the hydrodynamics to form realistic UFD stellar populations**, while following the evolution around **MW host halo** to $z=0$. This framework closes that gap with the following science goals:

- Formation threshold:** which subhalos host galaxies? *Occupation fraction* & *SHMR* of MW satellites.
- First stars:** *Pop III IMF* & *feedback* imprints on UFD *luminosities*, *sizes*, *chemistry*, and *SFHs*.
- Environment:** effect of *reionization*, *infall*, *stripping*, and host (MW vs. LMC) on quenching & structure.

Methods

To reach pc-scale grid resolution within a realistic cosmological volume, we combine:

ENZO & GRACKLE	AMR hydrodynamics & 9-species non-equilibrium chemistry.
Star formation (STARNET)	Pop III: (i) resolved physical criteria (ii) <i>StarNet</i> ML surrogate model + in-situ write. Pop II: H_2 -regulated + mechanical feedback.
Milky Way-est	Zoom-in ICs reproduce halo mass, mergers, & satellites.
Reionization	Lyman-Werner + UV background + MW radiation field.
Multi-region refinement	100 independent “must-refine” regions, one UFD each reaching. $\lesssim 3 \text{ pc}$, $\sim 10 M_\odot$.

TAB 2. Layers of the simulation framework. Dark blue is tested and light is next steps.

Future work

Our simulation will deliver:

- UFD Predictions:** occupation fraction + host-to-host scatter; stellar population, SMHM, quenching.
 - Mock observables:** photometry, luminosity functions, structure & kinematics, metallicity.
- In addition, we will reruns suites for robustness:
- Production suite:** 5 MW-est $\times$ 100 UFDs, $z \rightarrow 0$.
 - Reruns:** top-heavy Pop III IMF, weak-LW, elemental abundance, radiative transfer.

Acknowledgements: We thank Azton Wells, Will Hicks, Shin-Rong Tsai, and Britton Smith for useful discussions.
References: [1] Azton & Norman. ApJ 283.2:52. [2] Buch & Nadler et al. ApJ 971.1:79. [3] Wright et al. ApJ 970.1:70 [4] Tsai, Schive. & Turk. ApJS 284.1:30

